

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN
(Course Code: ECE 1003)

Module-2:Lecture-3

Combinational Circuit Design

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Module-2

- ❖ Binary adder and subtractor
- ❖ Ripple carry adders/subtractors and fast adders
- ❖ **Binary decoders**, encoders
- ❖ Multiplexers and de-multiplexers
- ❖ Logic functions using decoders and multiplexers
- ❖ Code converters
- ❖ Magnitude comparator

CONTENTS

Lecture-3

- ❖ Decoder
- ❖ Designing Higher order decoder for lower order decoder
- ❖ Boolean function Implementation using Decoder

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

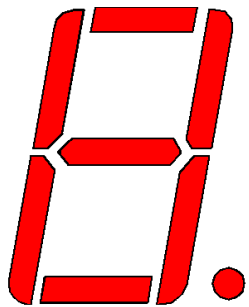
Decoder

DECODER

- ✓ In digital system, discrete information are represented by binary codes.

n bit code $\rightarrow 2^n$ discrete element

- ✓ A **decoder** is a combinational circuit that **converts** binary information of **n input lines to maximum of 2^n outputs** (may be fewer than 2^n outputs), i.e., n to m line decoder where **$m \leq 2^n$** . In general, the decoder take the form of **n x 2^n** . E.g. **2x4, 3x8, 4x16** and so on.
- ✓ **Decoders** can be used to **implement other combinational logics as well as Boolean functions**.
- ✓ E.g.: BCD to 7-segment decoder

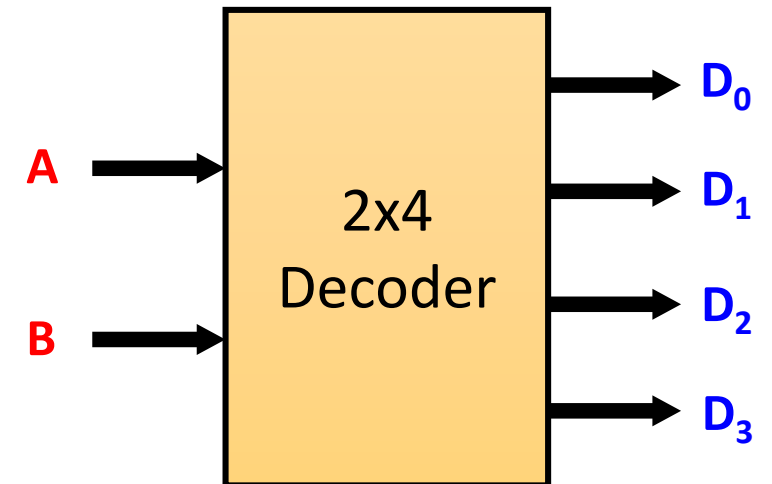


DECODER

2x4 Decoder

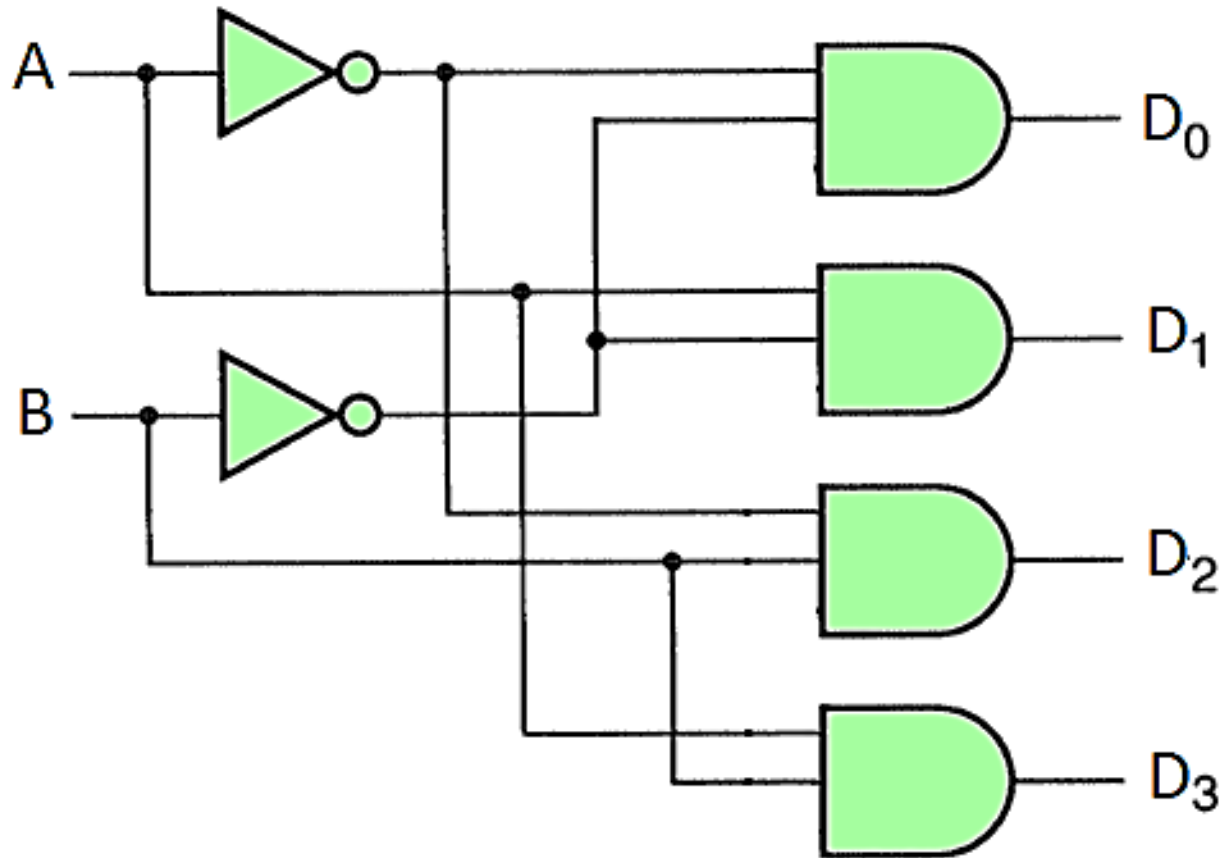
- ✓ A **2x4 decoder** has **2 inputs** and **$2^2=4$ outputs**.

A	B	D ₀	D ₁	D ₂	D ₃
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1



DECODER

2x4 Decoder



✓ Here,

$$D_0 = A'B'$$

$$D_1 = A'B$$

$$D_2 = AB'$$

$$D_3 = AB$$

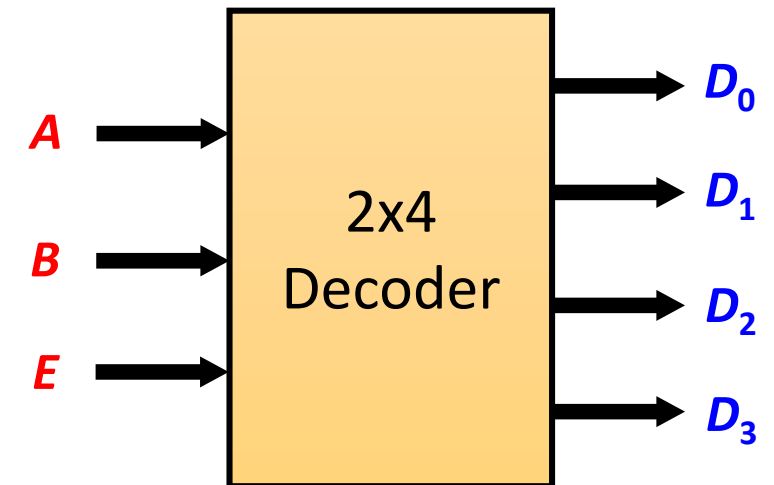
✓ Each represents one minterm

DECODER

2x4 Decoder with enable input

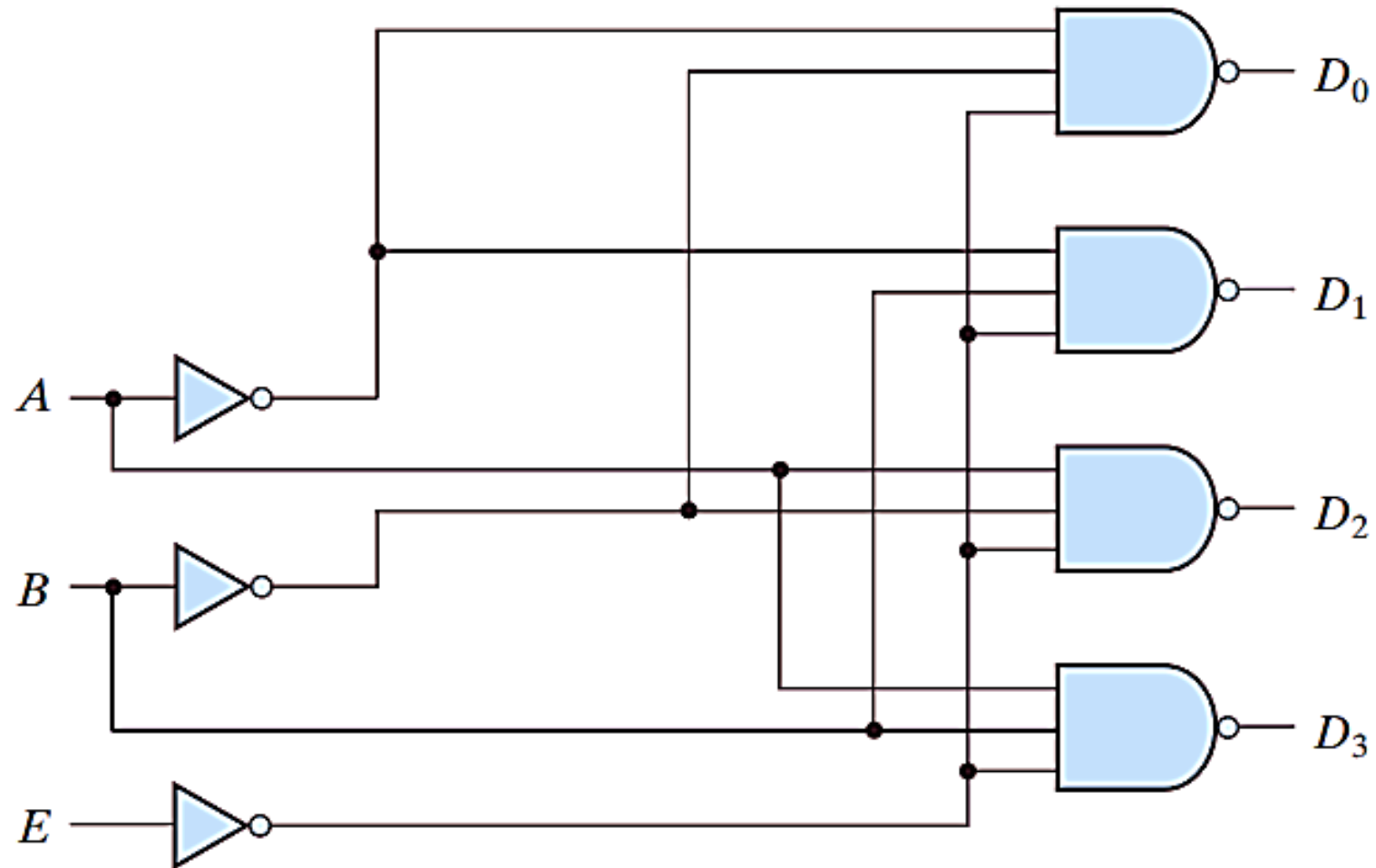
- ✓ An enable input used to enable the decoder operation. Here, the decoder is designed using NAND gates.

<i>E</i>	<i>A</i>	<i>B</i>	<i>D</i> ₀	<i>D</i> ₁	<i>D</i> ₂	<i>D</i> ₃
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0



DECODER

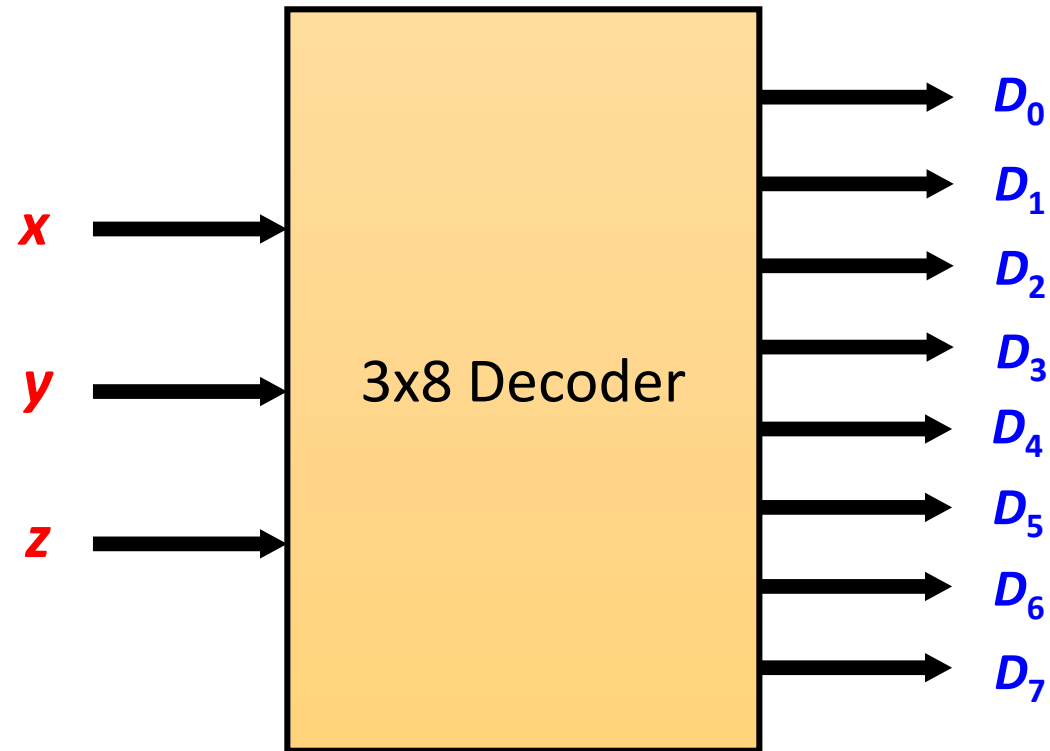
2x4 Decoder



DECODER

3x8 Decoder

- ✓ A 3x8 decoder has **3 inputs and $2^3 = 8$ outputs**.



DECODER

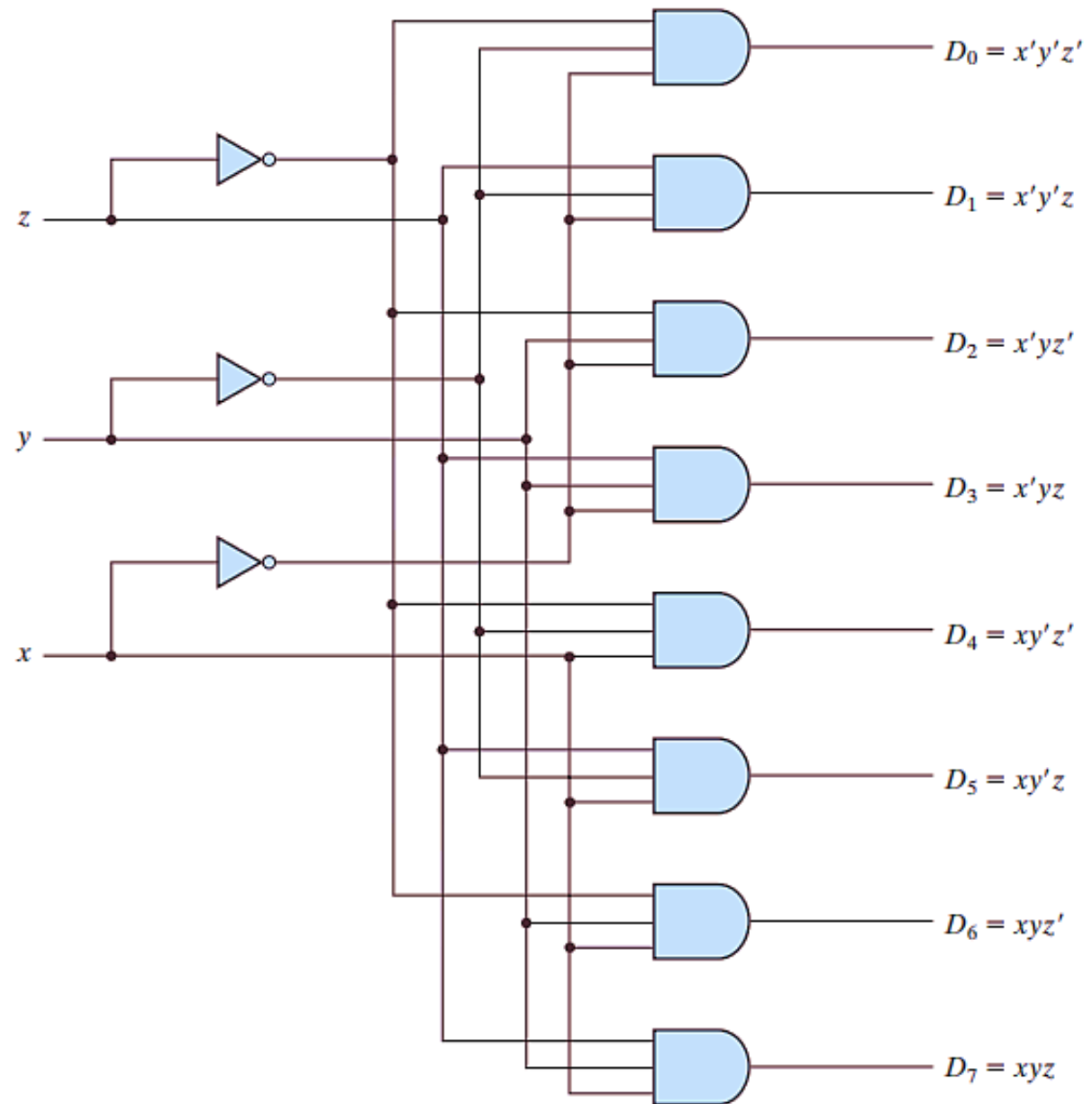
3x8 Decoder

Truth Table of a Three-to-Eight-Line Decoder

Inputs			Outputs							
<i>x</i>	<i>y</i>	<i>z</i>	<i>D</i> ₀	<i>D</i> ₁	<i>D</i> ₂	<i>D</i> ₃	<i>D</i> ₄	<i>D</i> ₅	<i>D</i> ₆	<i>D</i> ₇
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

DECODER

3x8 Decoder



DECODER

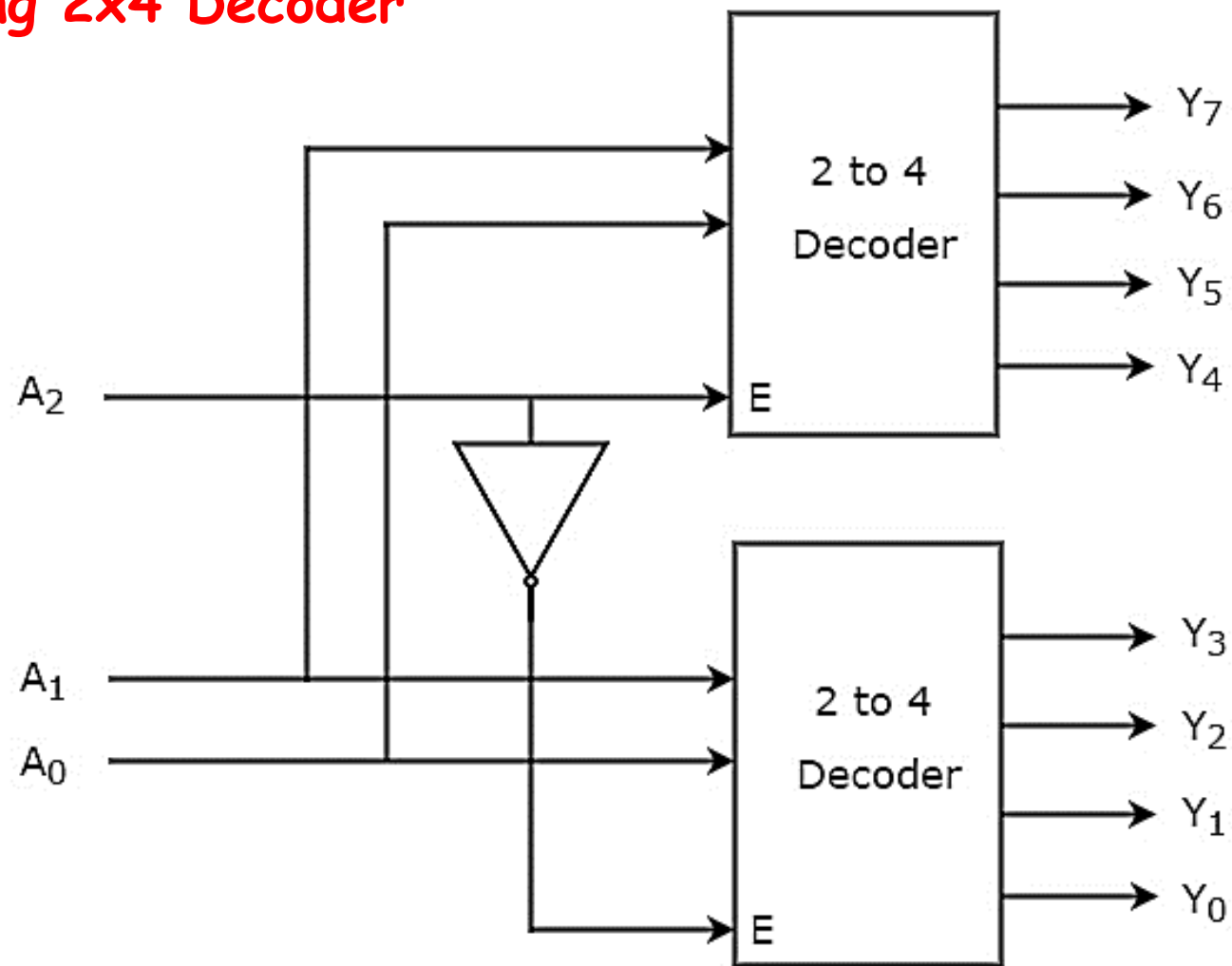
3x8 Decoder using 2x4 Decoder

- ✓ To design a 3x8 decoder, we require two 2x4 decoders.
- ✓ Here, both the 2x4 decoders will have same input A_0 and A_1
- ✓ A_2 can be connected to the enable pins of both the decoders

A_2	A_1	A_0	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

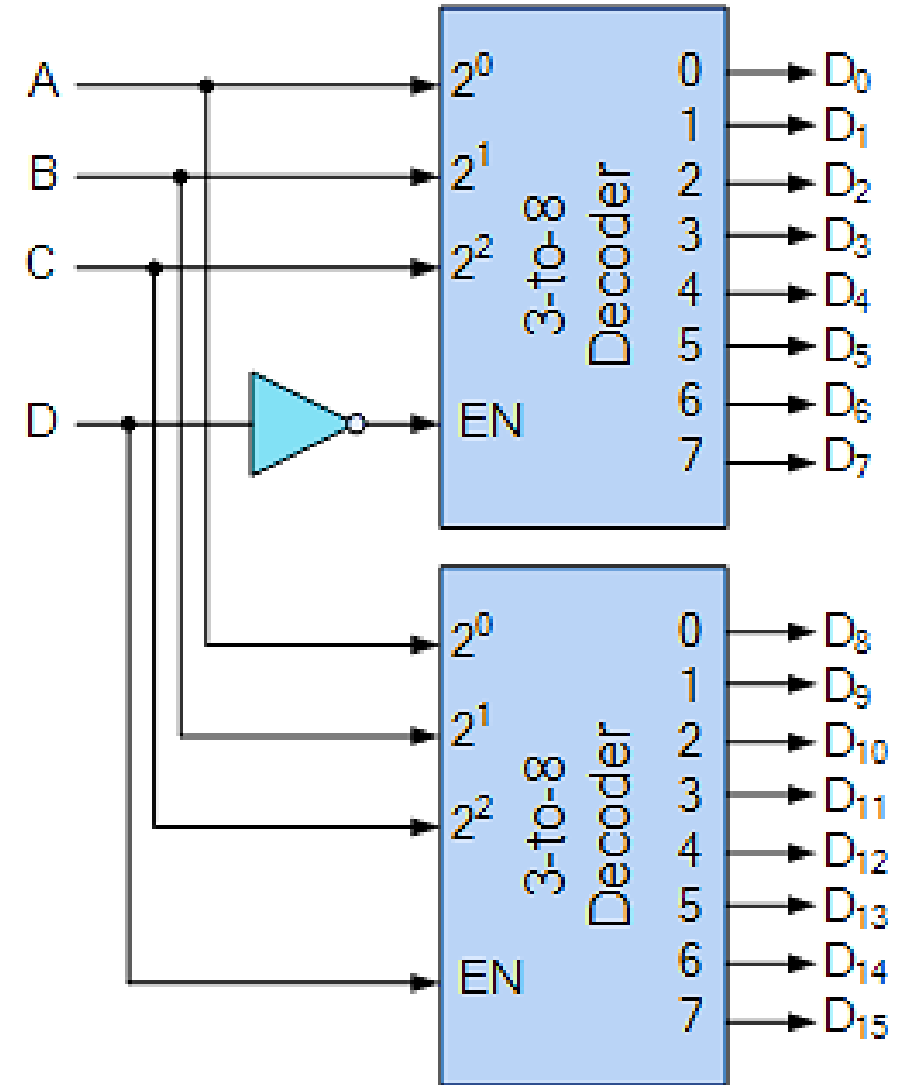
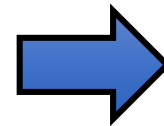
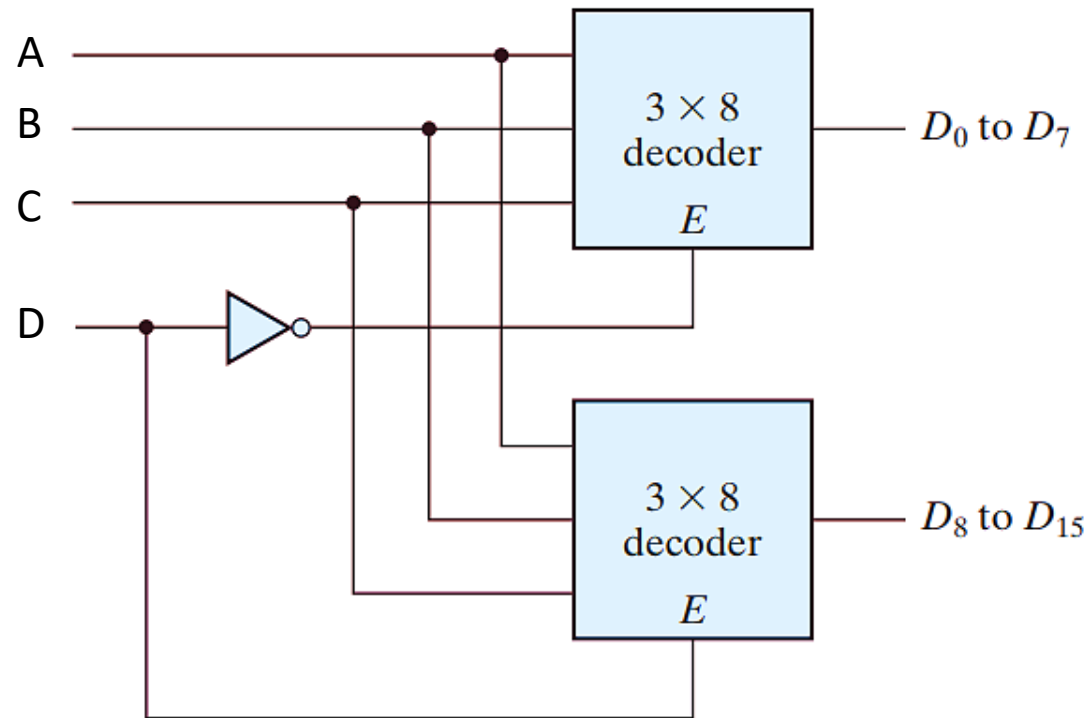
DECODER

3x8 Decoder using 2x4 Decoder



DECODER

4x16 Decoder using 3x8 Decoder



DECODER: PRACTICE PROBLEM

Question: Design a 4x16 decoder using appropriate number of

- (i) 3x8 Decoders
- (ii) 2x4 Decoders

DECODER

Question: Design the full adder circuit using appropriate number of decoders and logic gates.

Solution:

- ✓ From the truth table, it can be observed that

$$S(x, y, z) = \Sigma(1, 2, 4, 7)$$

$$C(x, y, z) = \Sigma(3, 5, 6, 7)$$

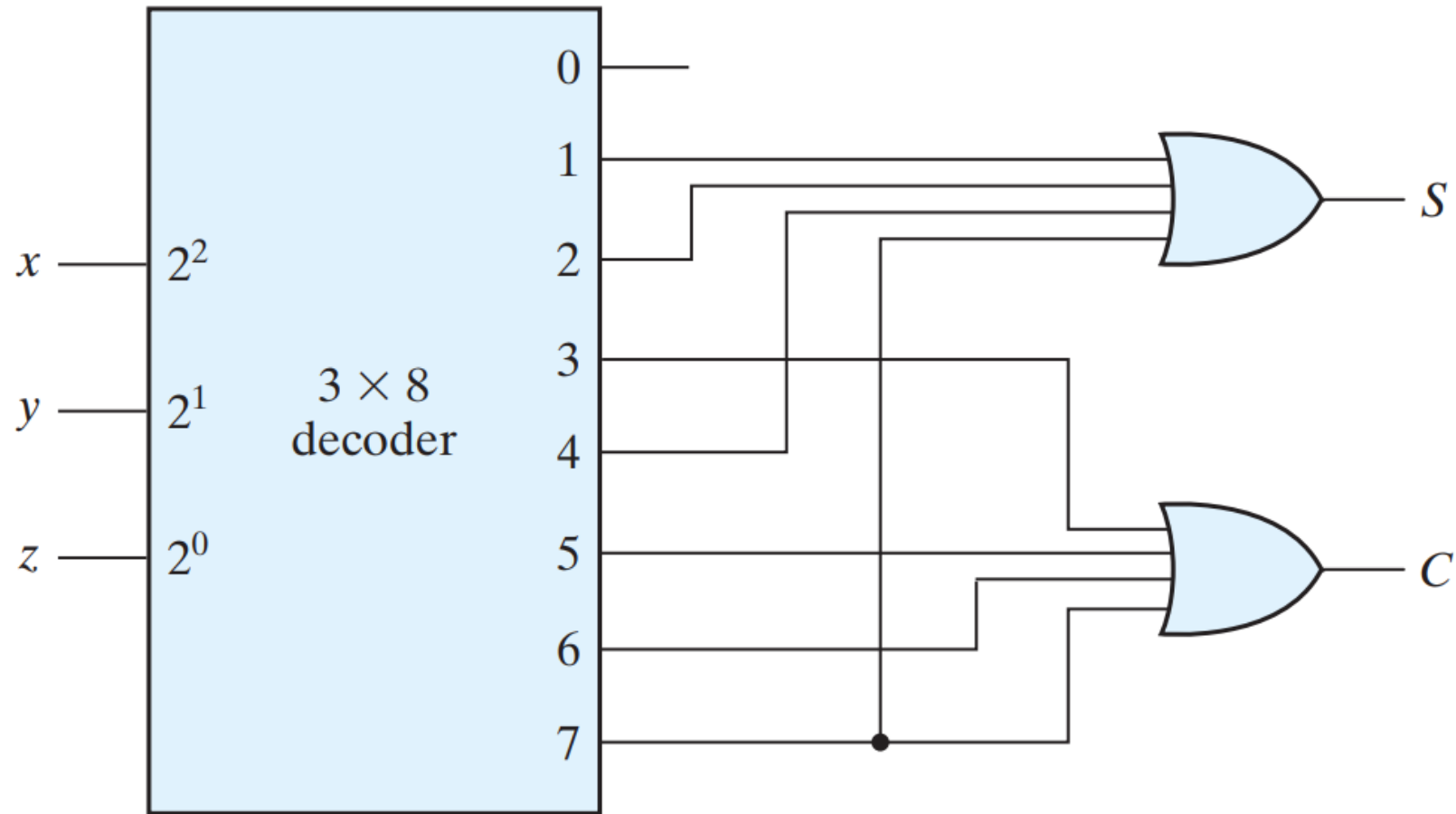
- ✓ Since there are three inputs and a total of eight minterms, we need a 3x8 decoder. The decoder generates the eight minterms for x, y, and z.
- ✓ The **OR gate** for **output S** forms the logical sum of minterms 1, 2, 4, and 7. The **OR gate** for **output C** forms the logical sum of minterms 3, 5, 6, and 7.

Full Adder

x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

DECODER

Logic Design:



DECODER

Question: Design the following Boolean function using appropriate decoder and logic gates

$$F = x'y + z$$

Solution:

$$F = x'y + z$$

$$= x'y(z + z') + (x + x')(y + y')z$$

$$= x'yz + x'yz' + (xy + x'y + xy' + x'y')z$$

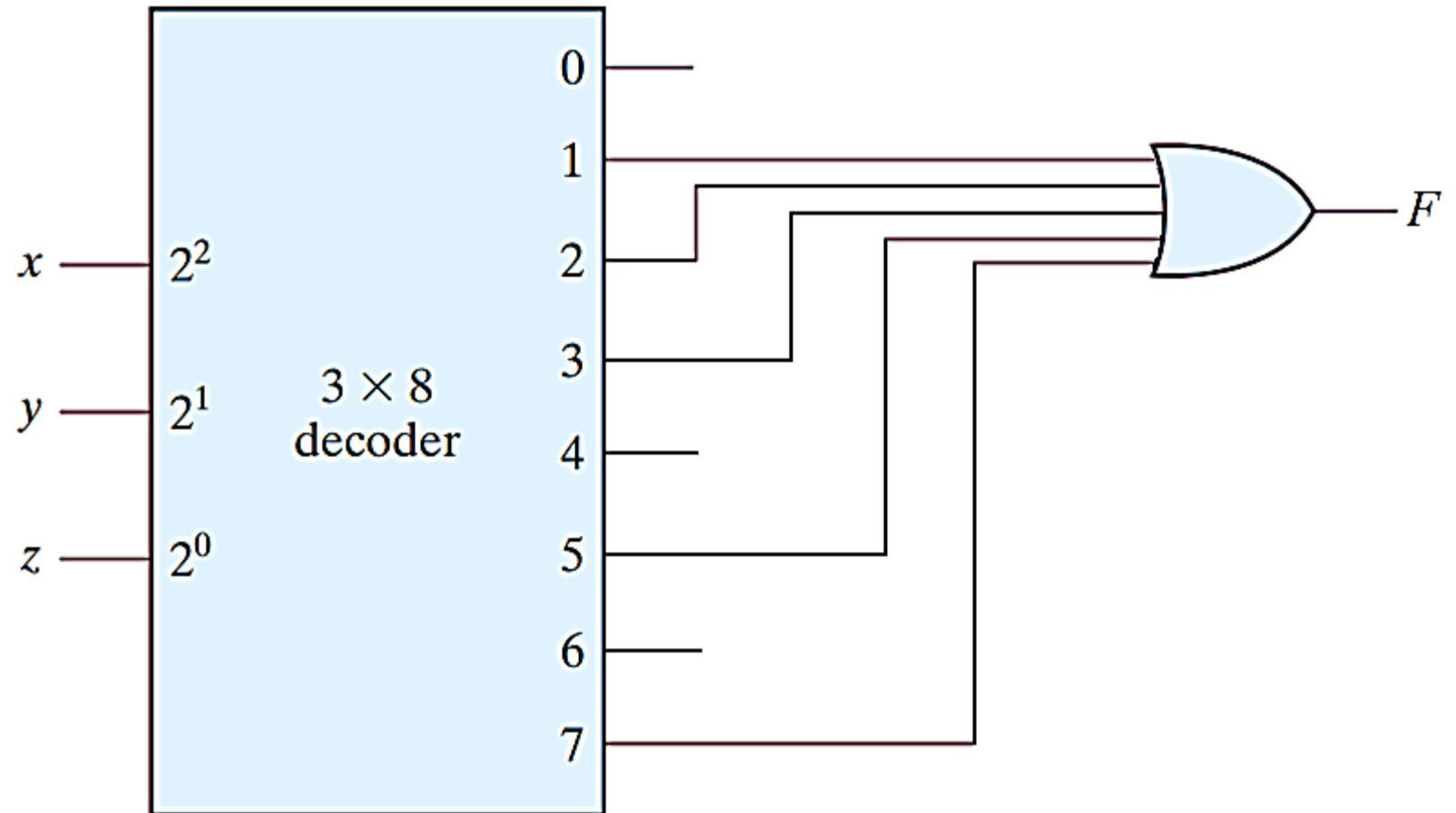
$$= \mathbf{x'yz} + x'yz' + xyz + \mathbf{x'yz} + xy'z + x'y'z$$

$$= \mathbf{m_3} + m_2 + m_7 + m_5 + m_1 = \mathbf{\Sigma(1, 2, 3, 5, 7)}$$

DECODER

Logic Design:

- ✓ As in this Boolean function has 3 input variables, so we require a 3x8 decoder and one OR gate for implementing the Boolean function.



Thank You